Anuj Pratap Singh Yadav

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SUMMARY

Staff engineer with 11 years of building large, long-lived C++ systems at Synopsys, Microsoft and Adobe. I own the coverage subsystems of a formal-verification product end to end, lead architecture changes across team boundaries, and build LLVM-based program analysis.

SKILLS

Languages: C++, Python, Tcl, C#, Java, TypeScript

Domain: VC Formal (DPV, FPV), formal verification, C++ and RTL code coverage, cone of influence, property density

Systems and tools: dependency inversion, IWYU, LLVM IR passes (dg), master/worker orchestration, serialization (XML, pugixml), Verdi, FlexNet, LSF, GDB, Linux, Perforce, Git, Win32

AI tooling: Cursor agent skills, rules and subagents built for the team's workflow, shared with the team

EXPERIENCE

R&D Staff Engineer | Synopsys *Noida, India | May 2023 - Present*

VC Formal (formal verification), DPV app: C++ vs RTL equivalence

- **Build-dependency decoupling (multi-org initiative)**: led VC Formal's part. The shared C++ layer under every app included engine headers, so one engine edit rebuilt every app. Cut wrong-way dependencies ~73% with include-what-you-use, hide-what-you-use (forward declarations), dependency inversion through startup-injected callbacks, and relocating misplaced code, rolled out in small reviewed changes to another team's code and shipped to production. Shared the design with the wider org in a tech talk.
- **C++ code coverage**: own it end to end. It shows which parts of a C++ model the formal proofs actually exercised, so C++ can be signed off like RTL. Added merging across proofs and export to the standard coverage database, so C++ and RTL coverage are reviewed together in one viewer; drive changes with the viewer and GUI teams, handle customer issues, and guided its integration into a new C++ + RTL sign-off app.
- **C++ program slicer (customer-requested R&D)**: building source-level cone of influence for C++ designs, which reports the exact C++ lines a property depends on. Built on the open-source dg LLVM slicer: wrote LLVM IR passes for C++ constructs it missed and a signal-based slicing criterion that cut whole-file slices down to the lines feeding each signal. Design reviewed weekly with a Distinguished Engineer.
- **Formal coverage (property density, RTL coverage)**: own both. They report which design signals, down to the bit, and which RTL lines the properties actually check. Rebuilt property density on the shared design model so DPV and FPV give the same answer, adding bit-level detail and RTL-to-RTL designs; aligned RTL coverage semantics across the two apps. In production.
- **Licensing for a distributed solver**: built token licensing in the C++ proof engine and moved runtime license checkout from workers to the orchestration master, so one process owns keys for every tier and each key serves more workers; fixed production checkout failures across license servers.
- **Session save/restore (serialization)**: extended snapshots to capture state that was silently lost on restore (user Tcl procedures, engine problem state as XML), and kept new features' state persistent with round-trip tests.

Software Engineer II | Microsoft *Noida, India | May 2021 - May 2023*

- Owned Recording Studio features in PowerPoint Win32 (C++): camera modes, live background blur, video export and Cameo, a camera object placed on the slide. Led v1 test automation and wired Recording and Export telemetry.
- Designed catalog filtering plus an allowlist (C#) so new commerce SKUs no longer broke OneDrive's upgrade path.
- Built the Stream 2.0 video playback component in Excel Win32 (C++, TypeScript) for SharePoint and OneDrive video.

Software Development Engineer 2 | Adobe *Noida, India | Aug 2015 - Apr 2021*

Software Engineer Aug 2015 - Jan 2017; SDE 2 from Jan 2017

- Moved Photoshop Elements and Premiere Elements from RIBS to Hyperdrive builds with parallel install bundles, cutting install time ~50% on average. Spot Award.
- Designed lazy loading for large files in the AEM web editor, cutting load time ~60% on files customers reported as freezing.

EDUCATION

B.Tech, Computer Science and Engineering - NIT Bhopal (MANIT) | 2011 - 2015 | GPA 8.27/10